
Humanoid Control Stack API Reference

Kevin Kawchak 
CEO ChemicalQDevice
10.5281/zenodo.xxxxxxxx

June 5, 2026

Abstract

A monospaced API reference for the humanoid control stack, from sensor-to-xyz through the planner to the broadcaster and safety checks.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Overview

This section introduces the control-stack codebase and states the scope of this reference. Replace this short paragraph with the opening context; one short paragraph per section keeps the template minimal and easy to extend.

2 Architecture

This section gives the background for the control-stack codebase. The single image placeholder below is based on the figure code of VVUQ-02 and renders a framed box until a real PNG is dropped into the Images folder.



Figure 1: Figure 1. Placeholder architecture diagram for the codebase.

3 Module Reference

This section records the control-stack codebase at a glance. The one table below is modeled on Table 1 of the VVUQ-05 final bill; keep cells short and ragged right so the table matches the body measure.

Work	What it is	Releases	Headline result	End-goal DOI
VVUQ-01	method and pipeline	v0.1.0	51 of 51 tests	xxxxxxx
VVUQ-02	ten-gate humanoid codebase	v1.0.0	172 of 172 tests	xxxxxxx

Table 1: Table 1. Module map behind this reference, at a glance.

4 Usage

This section analyzes the control-stack codebase. Keep the prose to one short paragraph and let a later revision expand each clause into its own section file.

5 Notes

This section concludes the treatment of the control-stack codebase and points to the back matter for acknowledgments, rights, citation, and data availability.

Acknowledgments

The author acknowledges Anthropic for access to Claude Code for the preparation of this reference; and OpenAI ChatGPT, Claude Haiku, and Google Gemini for research assistance.

Ethical Disclosures

The author declares no competing interests. This is an independent draft and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. All supporting numbers are simulation or illustrative results.

Rights and Permissions

This reference is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium provided the original author and source are credited. To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>. Reproduced public-domain text is used under 17 U.S.C. § 105.

Cite This Manual

Kawchak K. Humanoid Control Stack API Reference. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

Data Availability

All scaffolding, instructions, and supporting records are openly available in the kevinkawchak/cancer-automated and kevinkawchak/physical-ai-oncology-trials GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

- [1] K. Kawchak, “VVUQ Processing Priority over Code Generation: Method and Pipeline.” <https://doi.org/10.5281/zenodo.20372501>, 2026.
<https://doi.org/10.5281/zenodo.20372501>.
- [2] K. Kawchak, “Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ.” <https://doi.org/10.5281/zenodo.20421754>, 2026.
<https://doi.org/10.5281/zenodo.20421754>.
- [3] K. Kawchak, “Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510).” <https://doi.org/10.5281/zenodo.20535429>, 2026.
<https://doi.org/10.5281/zenodo.20535429>.

- [4] K. Kawchak, “National Physical AI Oncology Trial Platform.”
<https://doi.org/10.5281/zenodo.19244918>, 2026.
<https://doi.org/10.5281/zenodo.19244918>.
- [5] Creative Commons, “Attribution 4.0 International (CC BY 4.0).”
<https://creativecommons.org/licenses/by/4.0/>, 2013.
<https://creativecommons.org/licenses/by/4.0/>.
- [6] L. Lamport, *LaTeX: A Document Preparation System*. Addison-Wesley, 2nd ed., 1994.